

Voice Conversion 500-Pair

Out-of-Distribution Evaluation Report

EE 123 Project — Ved, Ching, Erick

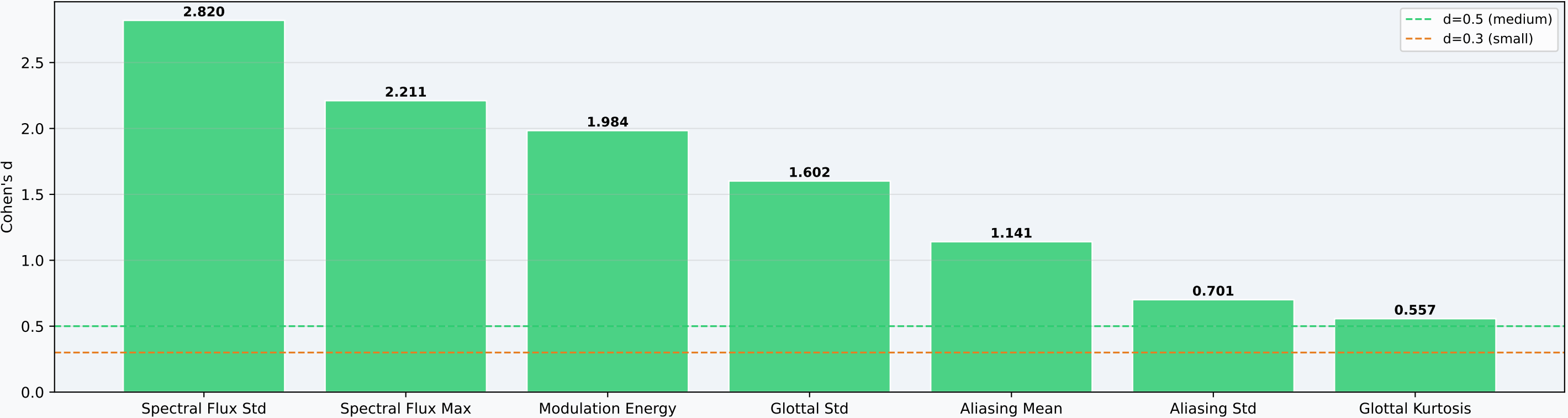
Real: 500 files (source-vctk500)

Spoof: 500 files (converted-vctk500)

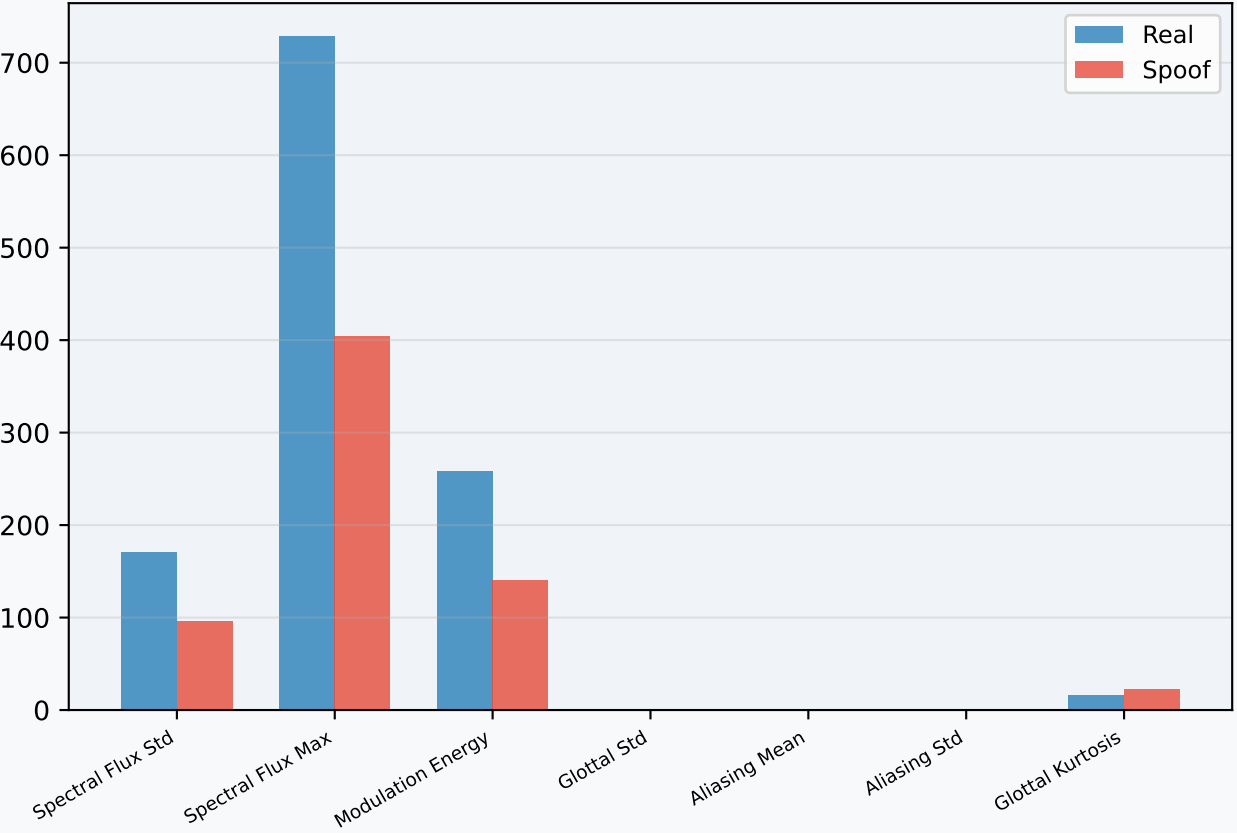
Models trained on VCTK 1000 vs 1000 (balanced)

Feature Discriminability — Voice Conversion 500 Pairs

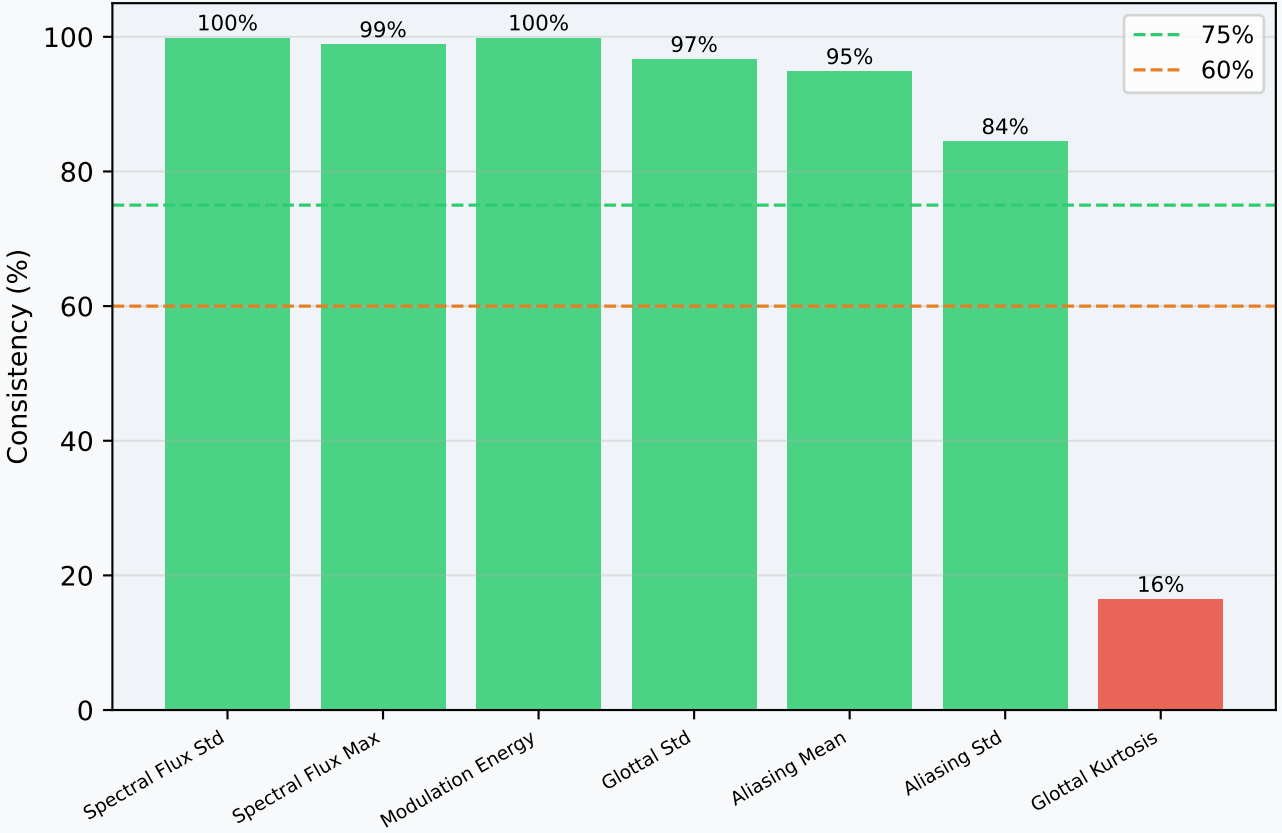
Cohen's d — Feature Discriminability



Mean Feature Values: Real vs Spoof



Consistency: % Pairs where Real > Spoof



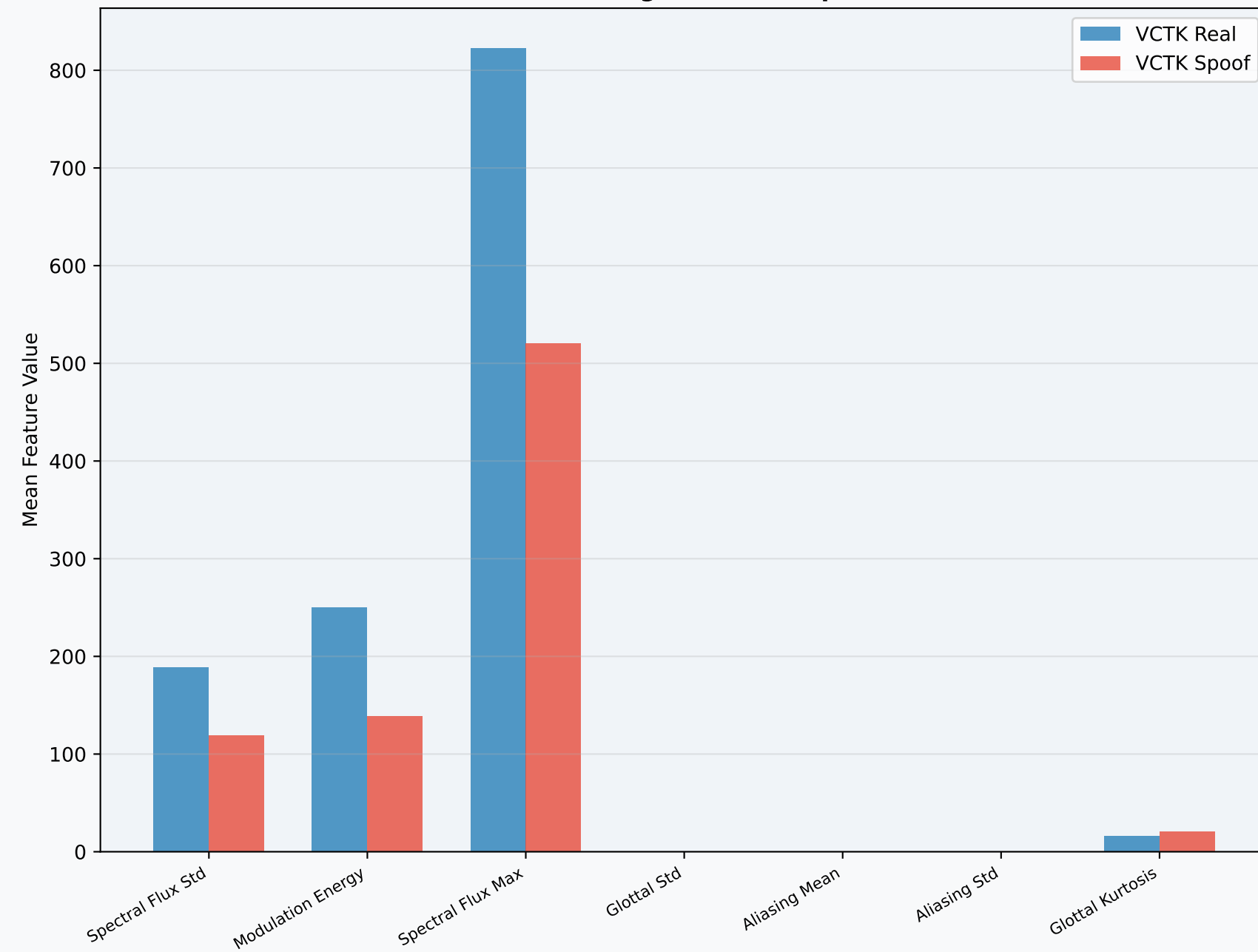
Voice Conversion 500-Pair Aggregate Feature Statistics

Feature	Real Mean	Real Std	Spoof Mean	Spoof Std	Diff%	Cohen's d	Real>Spoof (out of 500 pairs)	Consist%	Keep?
Spectral Flux Std	170.211	34.053	95.277	15.888	+44.0%	2.820	499/500	99.8%	✓ KEEP
Spectral Flux Max	727.999	186.407	404.259	90.178	+44.5%	2.211	494/500	98.8%	✓ KEEP
Modulation Energy	257.555	74.819	140.302	37.264	+45.5%	1.984	499/500	99.8%	✓ KEEP
Glottal Std	0.017	0.006	0.009	0.003	+45.3%	1.602	483/500	96.6%	✓ KEEP
Aliasing Mean	0.121	0.068	0.060	0.033	+50.4%	1.141	474/500	94.8%	✓ KEEP
Aliasing Std	0.160	0.106	0.101	0.056	+37.1%	0.701	422/500	84.4%	✓ KEEP
Glottal Kurtosis	15.956	8.793	22.337	13.602	-40.0%	0.557	82/500	16.4%	✗ DROP

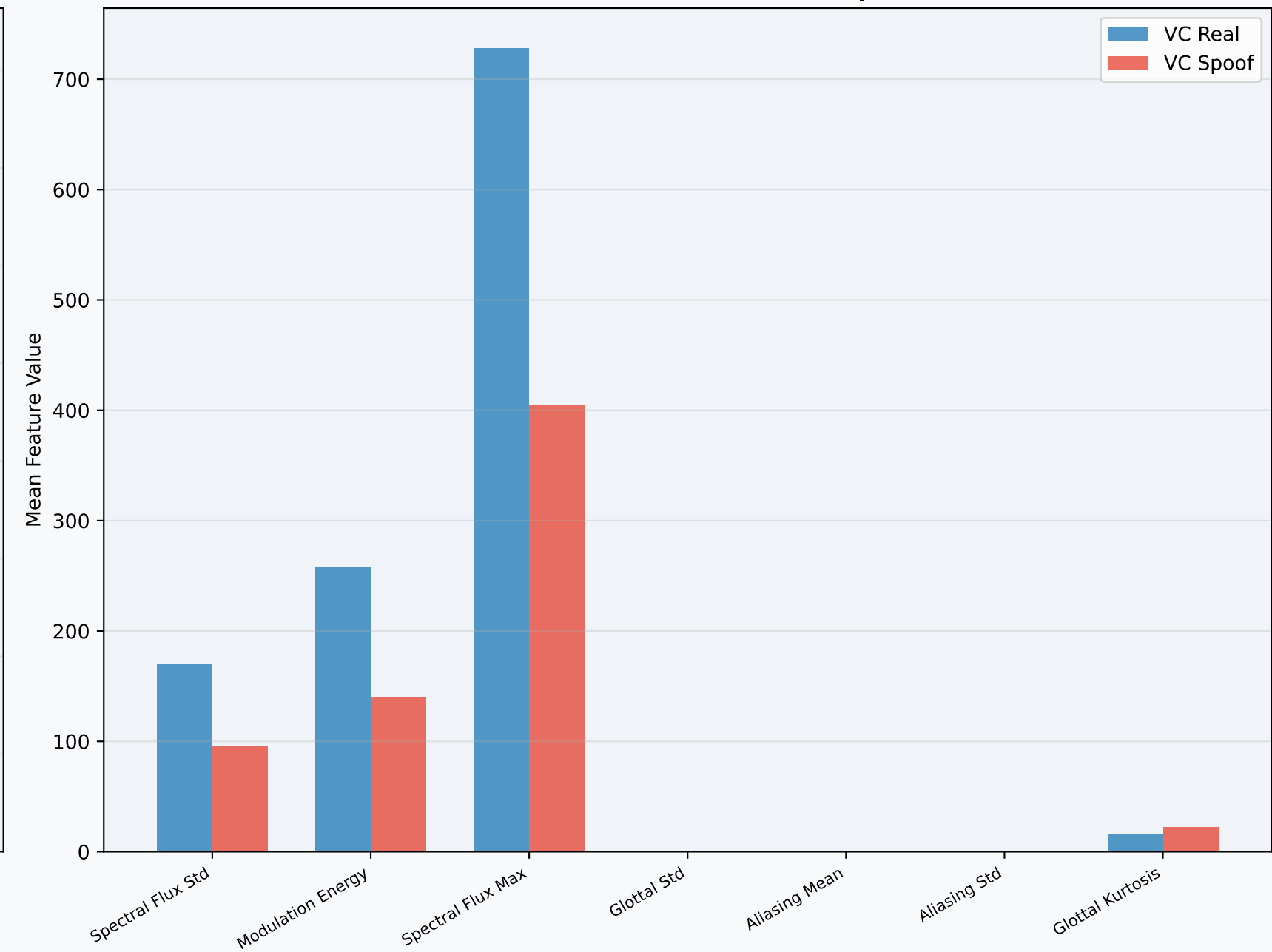
Green = strong (d>0.5) | Yellow = moderate (d>0.3) | Red = weak/drop

Feature Scale: VCTK Training vs Voice Conversion Dataset

VCTK Training Data (1000 pairs)

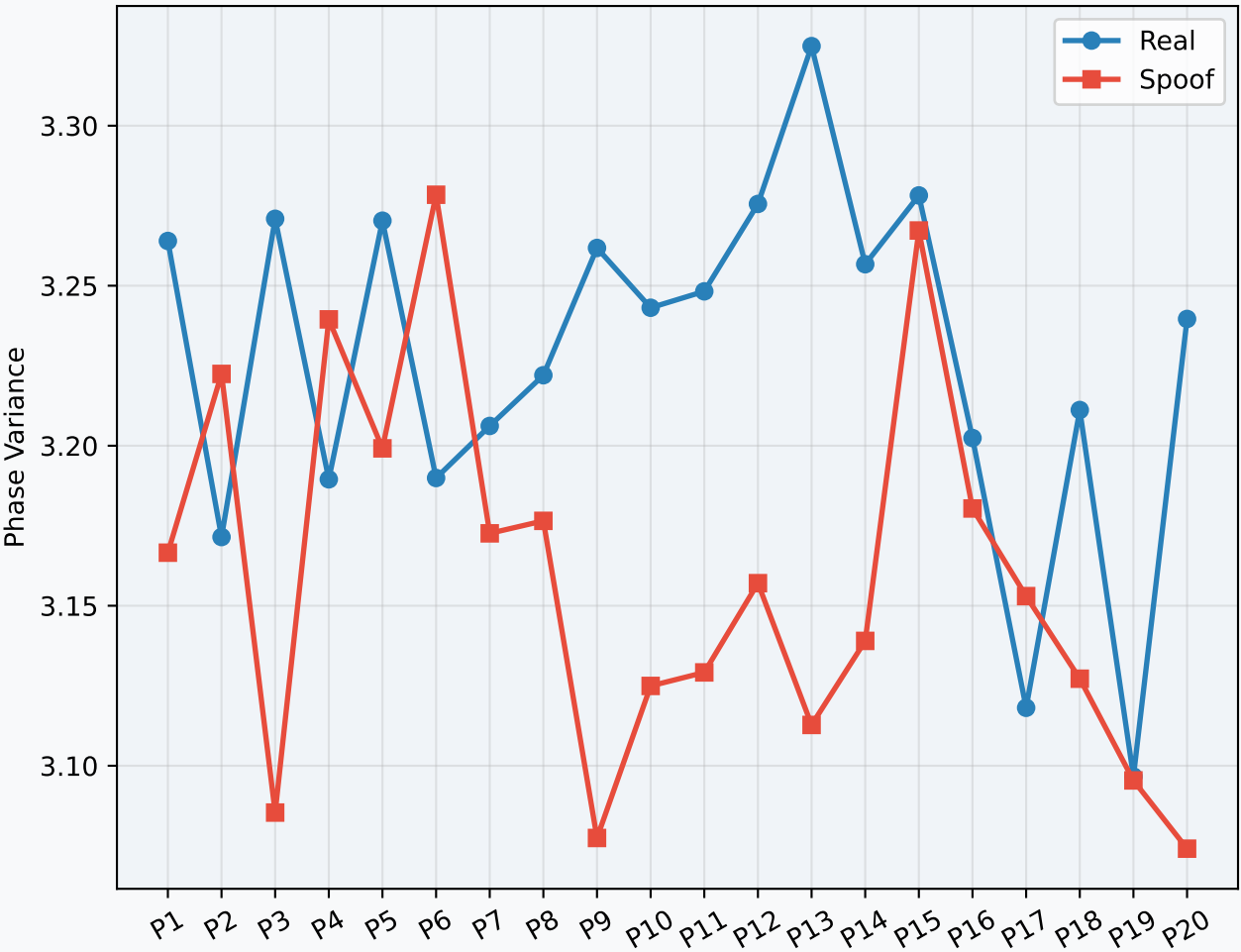


Voice Conversion Dataset (500 pairs)

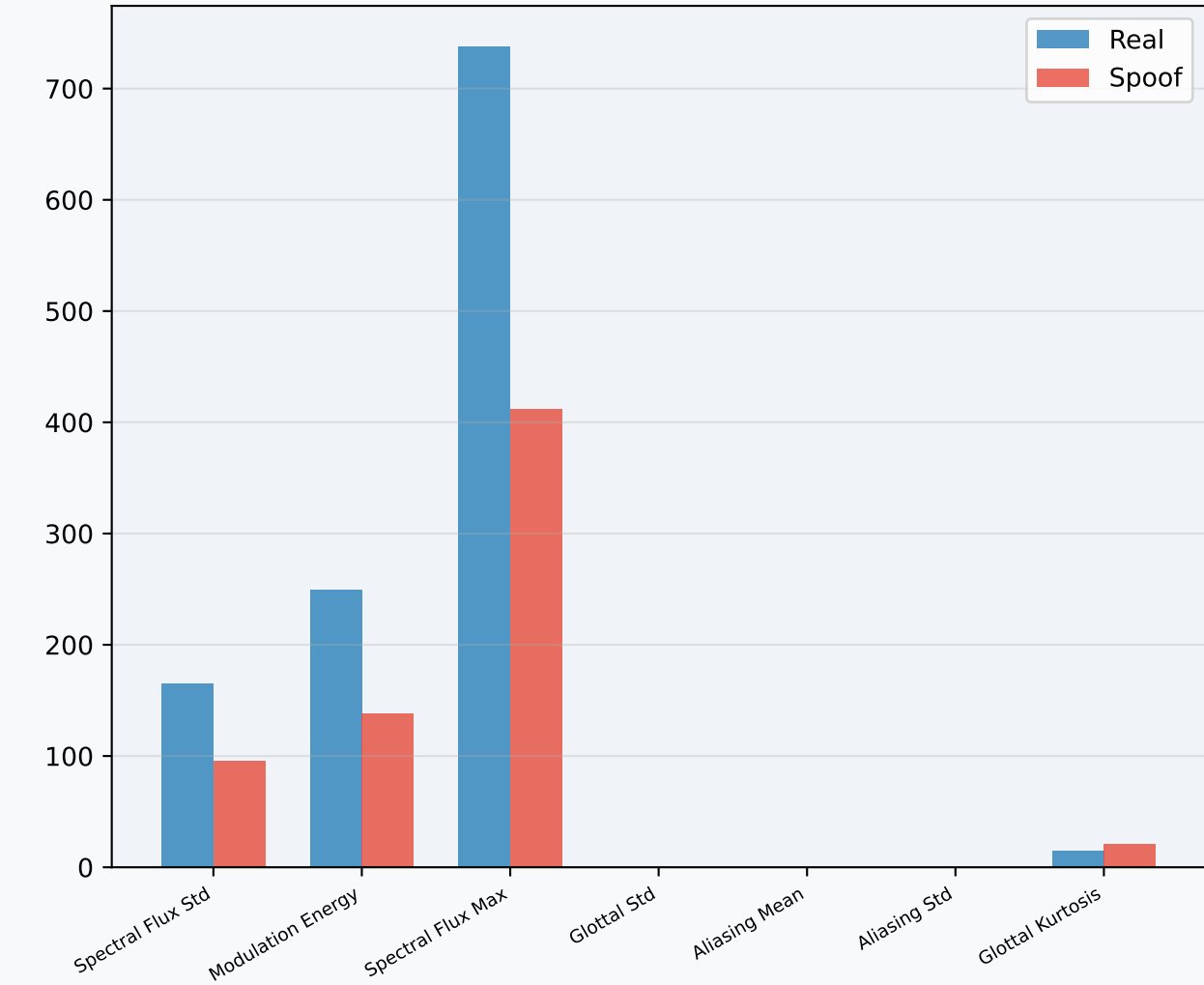


Signal Analysis Summary — 20 Sampled VC Pairs

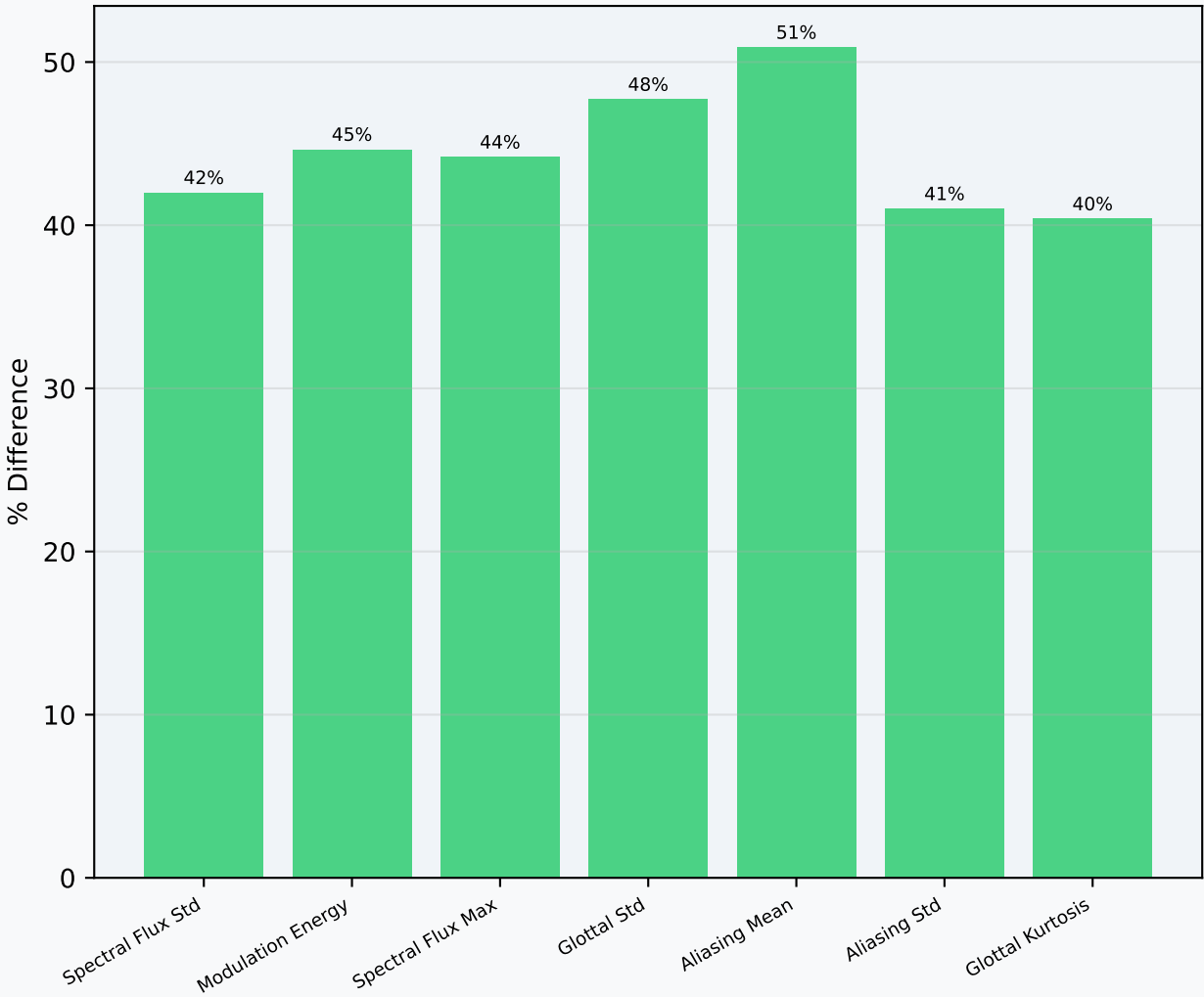
Mean Phase Variance per Pair



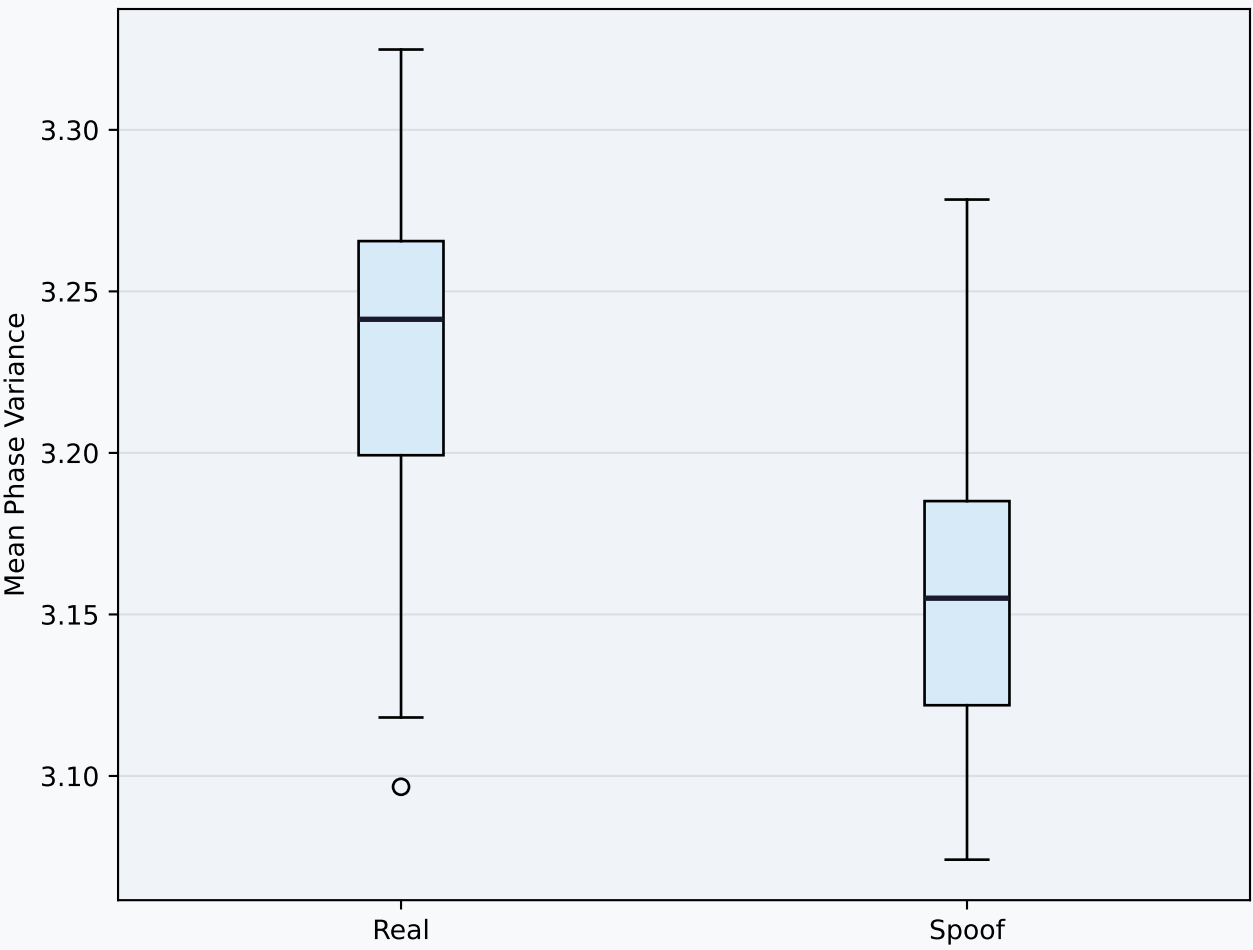
Avg Physics Features



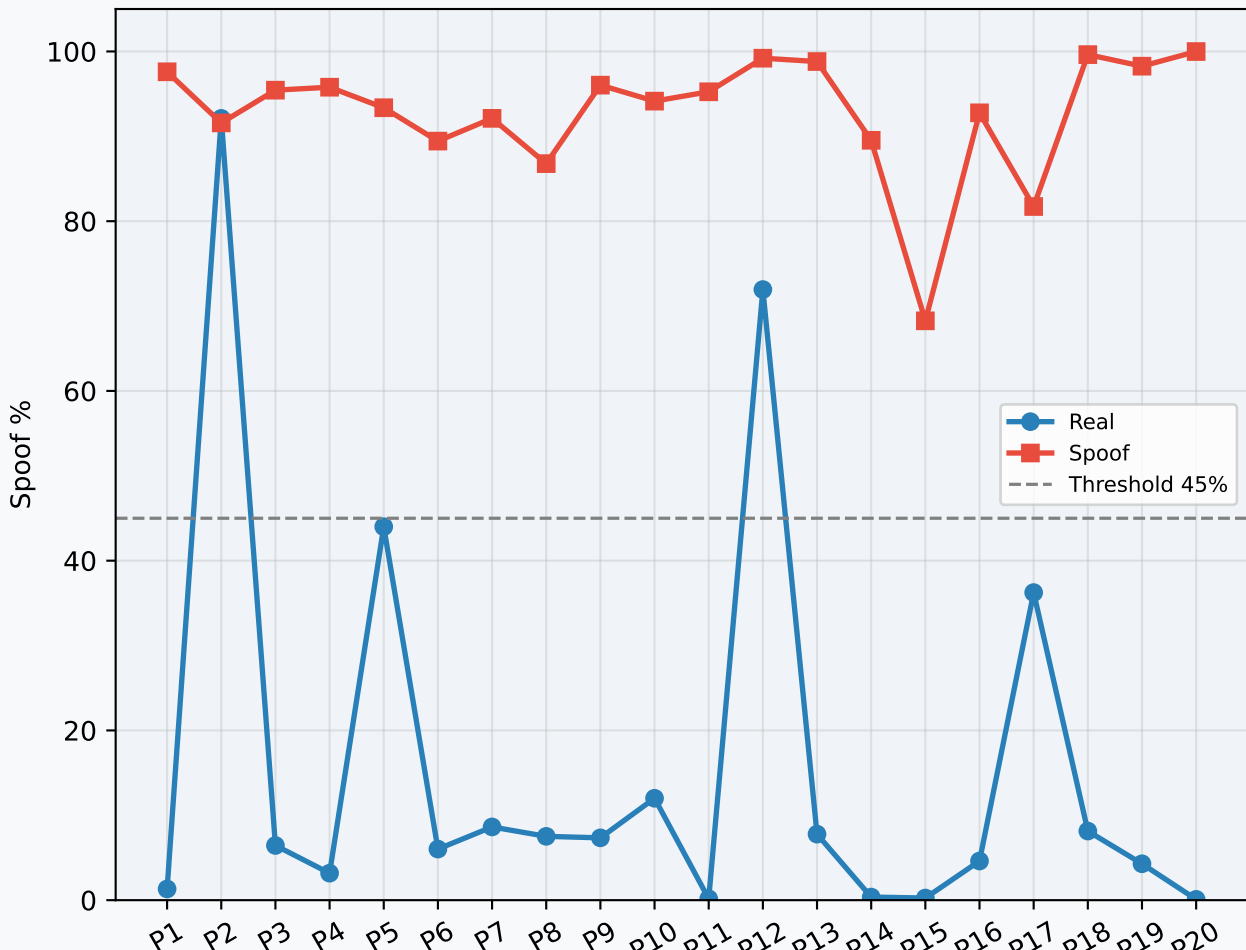
Avg % Difference per Feature



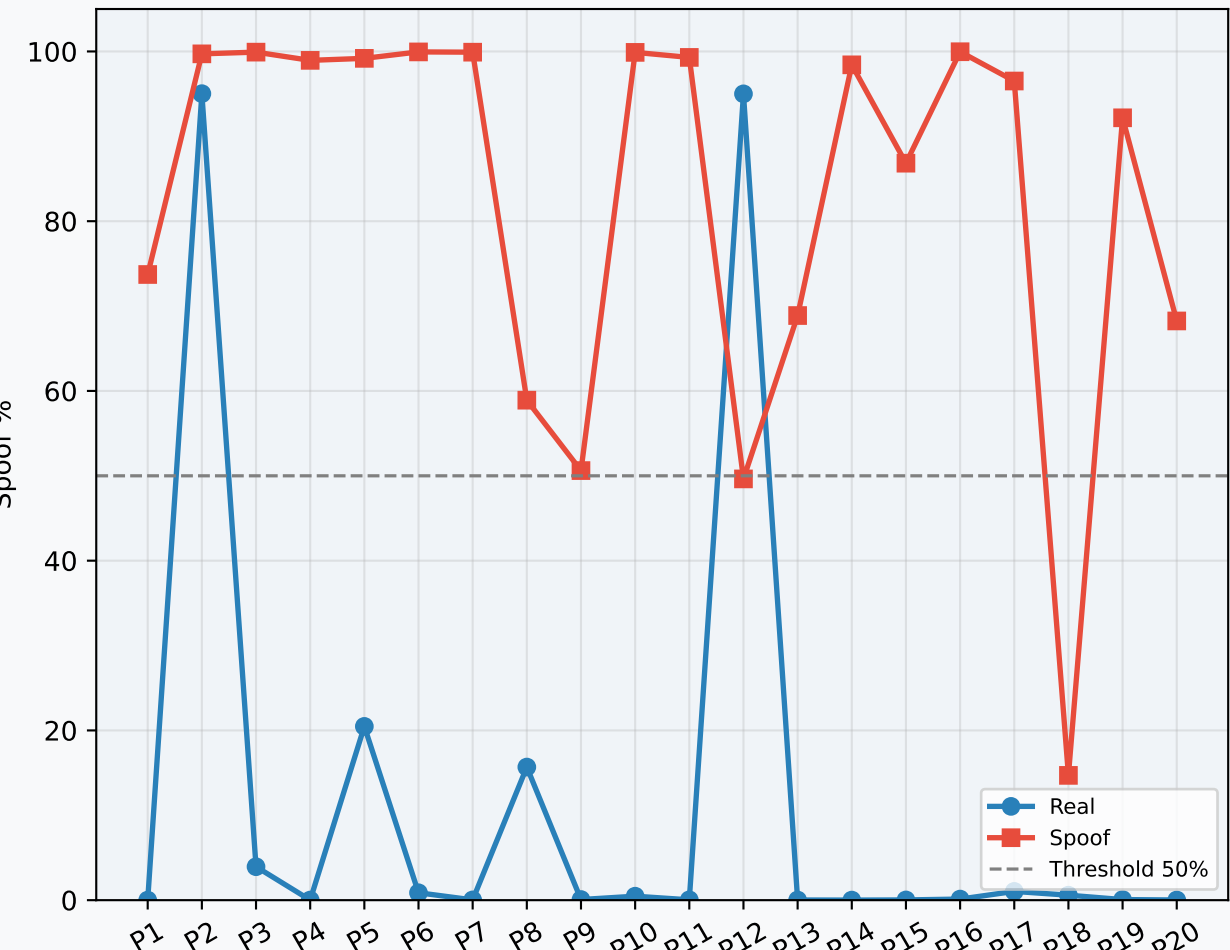
Phase Variance Distribution



LR Spoof Probability per Pair

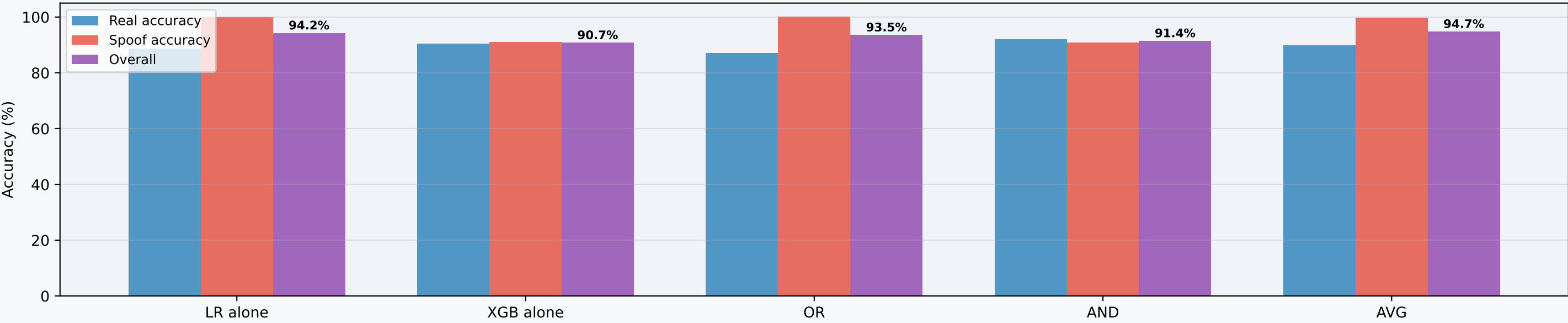


XGB Spoof Probability per Pair

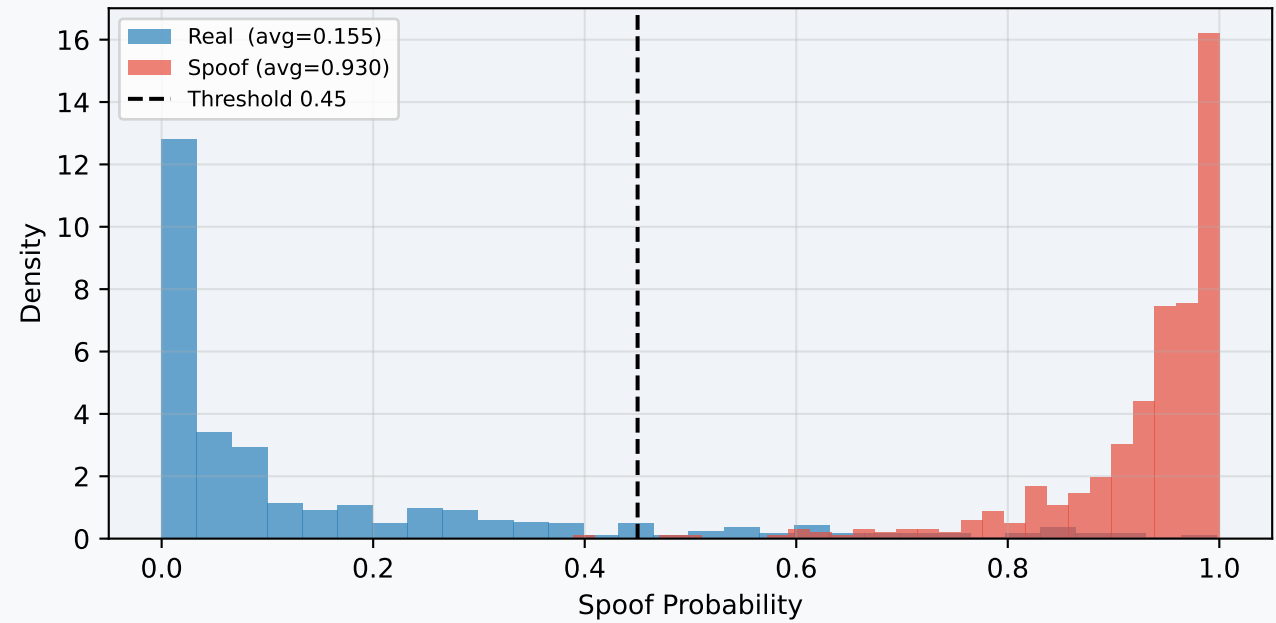


Voice Conversion 500-Pair — Ensemble Evaluation Results

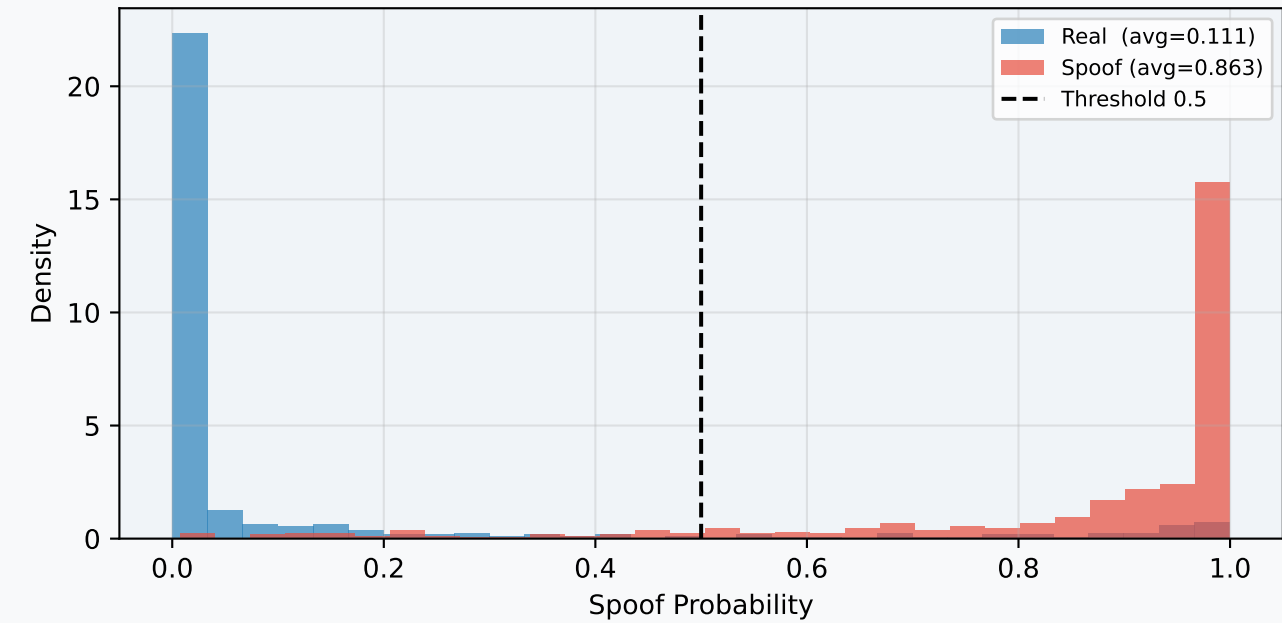
Accuracy by Strategy



LR Probability Distribution



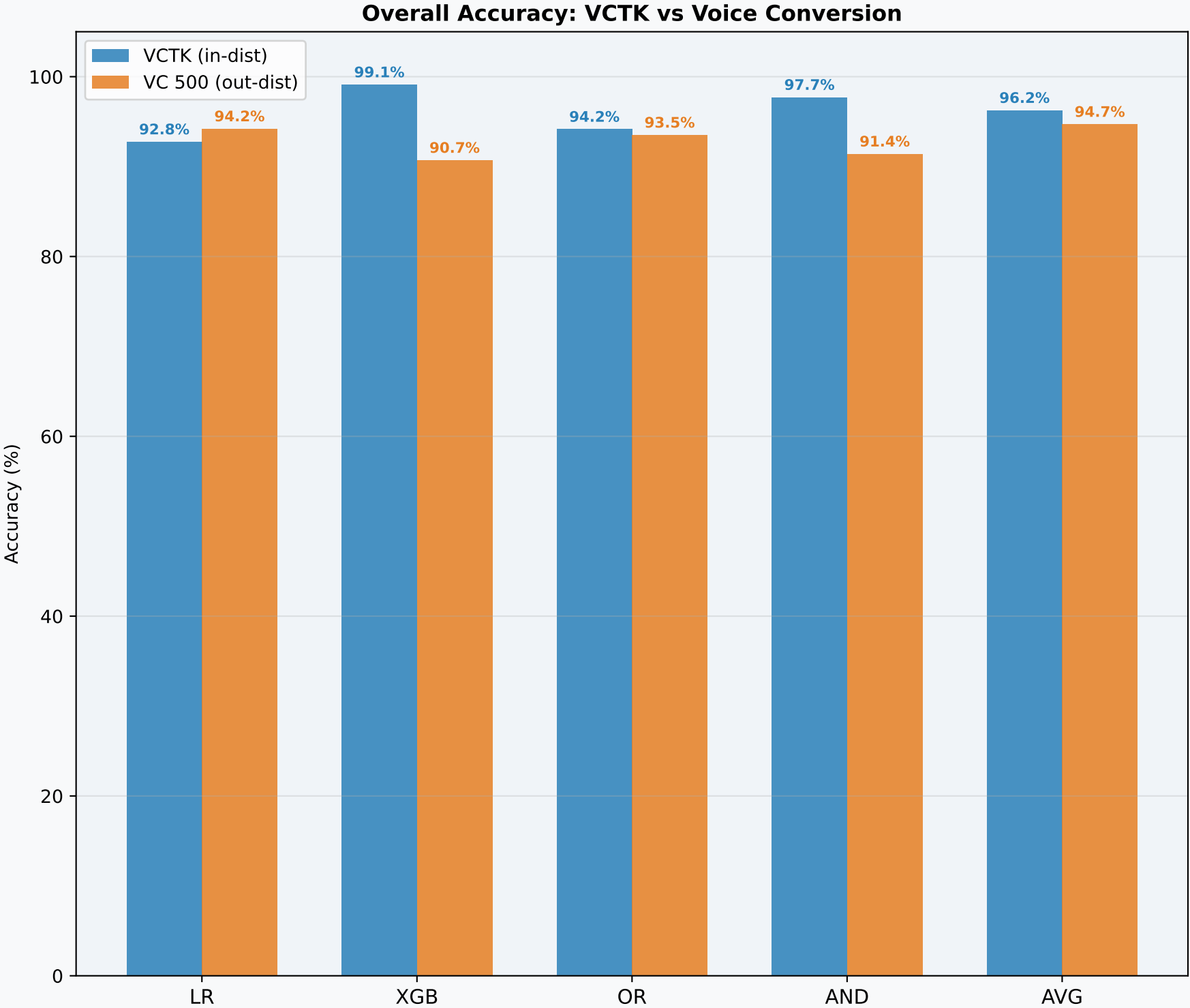
XGBoost Probability Distribution



Raw Count Summary

Strategy	Real Correct	Real Total	Real%	Spoof Correct	Spoof Total	Spoof%	Overall Correct	Overall Total	Overall%
LR alone	443	500	88.6%	499	500	99.8%	942	1000	94.2%
XGB alone	452	500	90.4%	455	500	91.0%	907	1000	90.7%
OR	435	500	87.0%	500	500	100.0%	935	1000	93.5%
AND	460	500	92.0%	454	500	90.8%	914	1000	91.4%
AVG	449	500	89.8%	498	500	99.6%	947	1000	94.7%

Cross-Dataset Comparison: VCTK Training vs Voice Conversion



Full Results Comparison Table

Method	Dataset	Real Correct	Real Tot	Real%	Spoof Correct	Spoof Tot	Spoof%	Overall
LR	VCTK 1k	885	1000	88.5%	970	1000	97.0%	92.8%
XGB	VCTK 1k	990	1000	99.0%	992	1000	99.2%	99.1%
OR	VCTK 1k	885	1000	88.5%	998	1000	99.8%	94.2%
AND	VCTK 1k	990	1000	99.0%	964	1000	96.4%	97.7%
AVG	VCTK 1k	929	1000	92.9%	996	1000	99.6%	96.2%
——	——	——	——	——	——	——	——	——
LR	VC 500	443	500	88.6%	499	500	99.8%	94.2%
XGB	VC 500	452	500	90.4%	455	500	91.0%	90.7%
OR	VC 500	435	500	87.0%	500	500	100.0%	93.5%
AND	VC 500	460	500	92.0%	454	500	90.8%	91.4%
AVG	VC 500	449	500	89.8%	498	500	99.6%	94.7%